

1. Why do we normalize data (centering values around zero) for neural networks? 1 point

 - ☐ To reduce the file size of the images.
 - ☒ To help the model train more effectively.
 - ☐ To convert images from color to grayscale.
 - ☐ To make the images look better when displayed.
2. Why should you call `model(input)` instead of `model.forward(input)`? 1 point

 - ☐ `model.forward()` doesn't exist—you can only use `model()`.
 - ☐ `model.forward()` only works during training, not evaluation.
 - ☐ `model()` runs faster than `model.forward()`.
 - ☒ Using `model()` handles PyTorch's internal setup properly.
3. Why does Mean Squared Error loss square the differences between predictions and targets? 1 point

 - ☒ To prevent cancellation and penalize larger errors.
 - ☐ To make all predictions fall between 0 and 1.
 - ☐ To convert negative predictions into positive values.
 - ☐ To make the loss function faster to compute.
4. Why do you evaluate your model on the test set after each training epoch, rather than just checking performance on the training set? 1 point

 - ☒ To see if the model learned to generalize.
 - ☐ Because the test set is larger and gives more accurate results.
 - ☐ Because the training set doesn't have labels, so you can't measure accuracy on it.
 - ☐ To update the model's weights based on test performance.
5. Which PyTorch code snippet correctly sets the GPU as the computing device if available? 1 point

 - ☐ `device = gpu()`
 - ☐ `device = torch.device('cpu' if torch.cpu.is_available() else 'gpu')`
 - ☐ `device = torch.set_device('gpu')`
 - ☒ `device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')`
6. Why is the Flatten layer needed before passing MNIST images to a Linear layer? 1 point

 - ☐ To apply the transforms like ToTensor and Normalize.
 - ☐ To reduce the size of the images to save memory.
 - ☐ To convert grayscale images to color images.
 - ☒ To reshape 2D images into 1D vectors.
7. Which optimizer is best described as a general-purpose optimizer suitable for various types of learning? 1 point

 - ☐ RMSProp.
 - ☒ Adam.
 - ☐ Stochastic Gradient Descent.
 - ☐ SGD with Momentum.
8. In the code walkthrough, the training DataLoader uses `shuffle=True`, but the test DataLoader uses `shuffle=False`. Why? 1 point

 - ☐ PyTorch automatically shuffles test data internally.
 - ☒ Training needs varied batches, but test order doesn't affect accuracy.
 - ☐ Test data must stay in order so you can compare predictions to the original dataset.
 - ☐ Shuffling is too slow for large test sets.
9. Why do you use `torch.no_grad()` during model evaluation? 1 point

- ☐ To switch the model from training mode to evaluation mode.
- ☐ To automatically calculate accuracy for the test set.
- ☒ To disable gradient tracking and save memory.
- ☐ To prevent the model from learning during evaluation.

10. What is the purpose of calling `model.eval()` before evaluating your model on test data?

1 point

- ☐ To change how certain layers behave during evaluation.
- ☒ To freeze the model's weights so they cannot be updated.
- ☐ To move the model from GPU to CPU for evaluation.
- ☐ To calculate the accuracy of the model on the test set.